



Head and neck injury in major trauma in Ireland: a multicentre retrospective analysis of patterns and surgical workload

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Abstract

Purpose Head and neck injuries are common in major trauma patients, but the prevalence is poorly documented in the literature. This study aims to investigate this further, particularly in the context of increasing development of Irish and European trauma systems.

Aims To determine the prevalence of, and patterns behind head and neck injury in major trauma in an Irish population.

Materials and methods The Trauma Audit Research Network database was analysed for major trauma patients (injury severity score > 15) with head and neck injuries admitted to any of the 26 participating hospitals in the Republic of Ireland between 2014 and 2017. The data was studied for patterns in the epidemiology of injuries and outcomes.

Results A total of 5364 patients were identified. Males were affected more than females (M:F 2.1:1).

The most common mechanism of injury was falls. There were 1272 counts of soft tissue injury, and 5814 counts of bony injury recorded within the cohort. Twenty-six percent of patients underwent some form of surgical procedure. Median length of hospital stay was 8 days, and the 30-day survival rate was 90%.

Conclusions A substantial number of Irish major trauma patients of all ages and genders sustain some form of head and neck injury. It produces significant injury and surgical workload involving input from multiple disciplines.

Keywords Epidemiology · Facial injuries · Head injuries · Multiple trauma · Western Europe

Introduction

Trauma remains a major public health problem. Each year, an estimated 5.8 million people globally die from it and many more are disabled. Trauma is predicted to become one of the top 20 leading causes of death worldwide by 2030. For people under 44 years of age, it is already among the top three causes

of death [1]. In Ireland, there were 5429 cases of major trauma diagnosed in 2018 alone [2].

Trauma of the head and neck comprises fracture of the facial bones, soft tissue injuries of the face and neck and dentoalveolar injuries [3]. A considerable number of injured patients sustain some form of head and neck trauma, either in isolation or in combination with other injuries [4]. Major trauma is any trauma that may cause death or severe disability. It can refer to a single injury or multiple injuries to the same or different body regions and systems [5]. Whether or not an injury is a major trauma can be determined using the injury severity score (ISS) [6]. An ISS of ≥ 16 is the cutoff for major trauma, and this is calculated by summing up the scores of the three most severe injuries a patient sustains. The management of head and neck trauma can be uniquely complex as it often requires input from a wide variety of surgical disciplines including maxillofacial surgery, plastic surgery and neurosurgery [7, 8].

Although head and neck injury in the general population has been well-studied to date [9–11], the literature has been relatively sparse with studies on it in a major trauma population [12, 13], and none in the Irish population.

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A greater understanding of this issue is important for the development of preventative initiatives aimed at reducing its impact not only on both individual morbidity and mortality but also on the health care system.

The Trauma Audit Research Network (TARN) is a European collaboration of hospitals including many in Ireland and the UK. It houses and maintains a database of major trauma patients who have fulfilled predetermined inclusion criteria regarding their injury, and the length and course of their hospital stay [14].

By analysing the TARN database, this study aims to:

1. determine the prevalence of head and neck injury in major trauma in an Irish population and
2. assess the sociodemographic and clinical patterns underlying major trauma in this population

Materials and methods

The TARN database was searched for major trauma patients with head and neck injuries admitted to any of the 26 participating hospitals in the Republic of Ireland between 2014 and 2017 (inclusive).

The TARN database collects data on over 100 variables relating to the patient, their injury and hospital stay. For the purposes of this study, a selection of relevant variables was chosen for analysis and was divided into two groups:

The first included sociodemographic measures relating to the patient cohort as well as the circumstances and factors relating to their injuries, management and outcomes. These included age, sex, injury location, evidence of intoxication on admission, and injury intent.

The second group consisted of clinical variables. These were the mechanism of injury, whether it was blunt or penetrating, the anatomical location of the injuries themselves (further subdivided into soft tissue and hard tissue injuries), injury severity, treatment received, length of stay in hospital, and 30-day mortality rates.

Descriptive analysis was performed using SPSS (IBM SPSS® Statistics for Macintosh, Version 26).

This study received ethical approval from the Clinical Research Ethics Committee of University College Cork.

Results

A total of 19,483 major trauma patients fulfilling the criteria for inclusion in the TARN database were admitted to one of the 26 TARN-affiliated hospitals across Ireland in the 4 years between January 1, 2014, and December 31, 2017. Of these,

5364 (27.5%) sustained some form of head, face and/or neck injury.

Age and sex

The mean age of the sample was 53.57 years (± 25.3 SD, range 0.05–102.4). However, 2155 patients (40%) were over the age of 65. Figure 1 shows this distribution in deciles. The 70–80 decile had the highest number of patients at 879 (16%).

Within the sample, 68% of patients were male (3653) and 32% were female (1711) giving a male:female ratio of 2.125:1.

Injury location

The home by far is the most frequent place of injury, comprising 43% (2322) of the sample. Roads and public areas were the next most frequent, at 22% (1177) and 20% (1079) respectively. In 119 cases (2%), the injury location was not known. The office was the least common place to be injured, with only 4 patients (< 1%). Other locations of injuries included tube tracks and rail tracks. This data is clarified in Table 1.

Intoxication

Evidence of intoxication was described in 1602 cases (29.8%). Of these, 95% were alcohol-related, with the remaining 5% (95) cases being related to drugs, or both alcohol and drugs.

Injury intent

As shown in Table 2, the vast majority of patients sustained their injuries non-intentionally. Suspected self-harm or where the intent was inconclusive represented a minority of cases, which amounted to 6% (288) of the cohort.

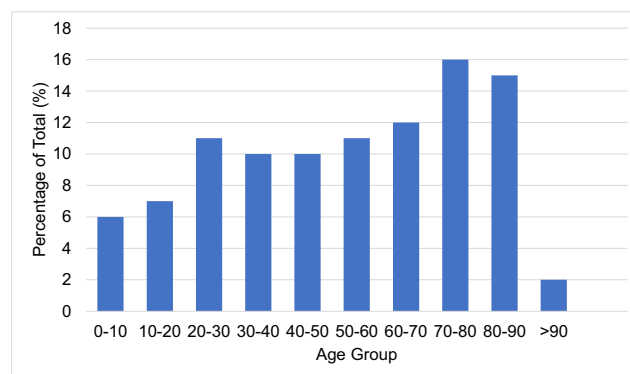


Fig. 1 Percentage age distribution by decile

Table 1 Distribution of injury location

Place of injury	No. of persons	% of persons
Home	2322	43
Road	1177	22
Public area	1079	20
Institution	262	4
Farm	183	3
Not known	119	2
Industrial	97	2
Other	83	1.5
Mountain	23	< 1
Water	15	< 1
Office	4	< 1
Total	5364	100

Mechanism and nature of injury

Falls of some description were the most frequent aetiology leading to injury in the cohort, accounting for approximately two-thirds of cases (3558 patients). The least common causes included stabbings/shootings, crush injuries, burns and blasts, each accounting for less than 1%. Table 3 illustrates the different aetiologies of the trauma sustained by the cohort as a whole.

The major aetiology of injury did vary somewhat with age. This is depicted in Fig. 2 with the sample being divided into 3 age groups. The most common aetiology remained falls in the 35–70 and > 70 groups. However, in the 0–35 age group, road traffic accidents (RTAs) were the most frequent aetiology, decreasing in frequency with increasing age. While blows represented 10% (552) of the whole sample, over half of these were among patients in the 0–35 group (297). Ninety-nine percent of injuries were blunt in nature (5289), with 1% (75) being penetrating.

Injuries - soft tissue

There were a total of 1272 counts of soft tissue injuries within the sample. The majority of these were injuries sustained to

Table 2 Distribution of injury intent

Injury intent	No. of persons	% of persons
Non-intentional	4528	84
Alleged assault	399	7
Intent inconclusive	221	4
Sport	149	3
Suspected self-harm	67	2
Total	5364	100

Table 3 Distribution of mechanism of injury

Mechanism of injury	No. of persons	% of persons
Fall < 2 m	2721	51
Road traffic accident	1130	21
Fall > 2 m	837	16
Blow	552	10
Other	60	1
Stabbing/shooting	44	< 1
Crush	12	< 1
Other	83	1.5
Burn	5	< 1
Blast	3	< 1
Total	5364	100

the scalp or the face (Fig. 3). Scalp contusions were the most frequently recorded injury (39%). Conversely, vascular injuries were the least frequently encountered (4%).

Injuries - hard tissue

There were a total of 5814 hard tissue injuries, comprising fractures of the bones of the head, face and neck. There were more injuries than number of patients as patients often sustained more than one injury. Skull fractures were the most common (55%), followed by cervical spine fractures (16%). Fractures of the various components of the facial skeleton were the least frequently seen hard tissue injuries, and a description can be seen in Fig. 4.

Injury severity

Of the injuries included in the sample, 4874 (81%) were head injuries, and 1134 (19%) were facial injuries. Neck injuries were excluded from injury severity analysis as they lack their own dedicated Abbreviated Injury Scale (AIS) scores.

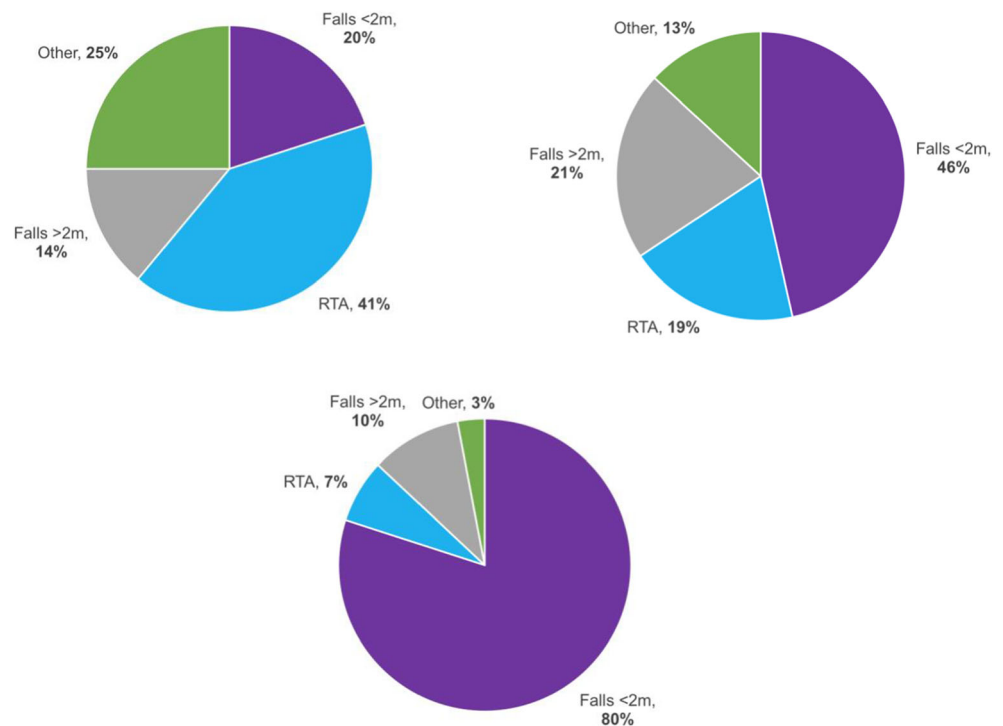
Figure 5 demonstrates injury severity for head (left) and facial (right) injuries according to the 6 grades of the AIS. All facial injuries were graded 1–3, and no facial injuries in the sample were grade 4 or higher.

Management

Surgical intervention was performed in 26% (1402) cases. Of these, 320 (6%) subsequently underwent a second procedure, and 115 (2%) underwent a third.

There were a total of 1837 individual operations recorded. Neurosurgery was the most frequently performed, while maxillofacial and plastic surgery were the 4th and 5th most commonly performed respectively.

Fig. 2 Mechanism of injury by age group (0–35 (top-left), 35–70 (top-right), > 70 (bottom))



Outcomes

The 30-day mortality rate was 10% (539) with 4825 (90%) of patients surviving their trauma and ensuing hospital stay.

Median length of stay was 8 days. Table 4 shows the length of stay distribution across the sample.

Discussion

Major trauma places a high burden on health care systems [15]. Head and neck injury in particular often results in complex injuries which in turn produce both considerable financial [16] and personal impact [17].

The present study was carried out to establish the prevalence of head, face and neck injury in the setting of an Irish

major trauma population. It sought to examine the sociodemographic (age, sex, etc.) and clinical factors (injury types, treatments) underlying it. The data were extracted from one of Europe's largest major trauma databases, the TARN database.

Data from 26 participating hospitals across the republic of Ireland between January 2014 and December 2017 were analysed producing a study population of 3653 males and 1711 females.

Out of the 19,483 major trauma patients included in the TARN database during the study period, 5364 sustained some form of head and neck injury, giving a prevalence of 27.5%. This is higher than that reported in other studies carried out with comparable methodologies and over similar time frames which showed rates of 16% [12] and 17% [13].

However, these studies included maxillofacial injuries only, and therefore may not have included patients with head or

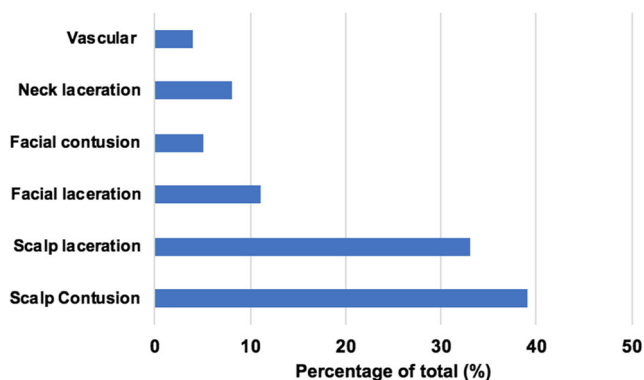


Fig. 3 Distribution of soft tissue injuries by type

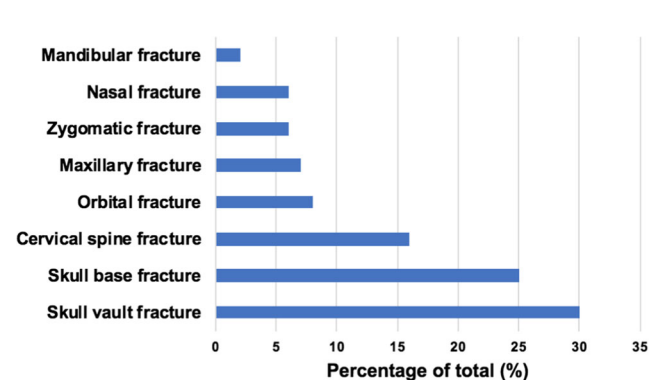
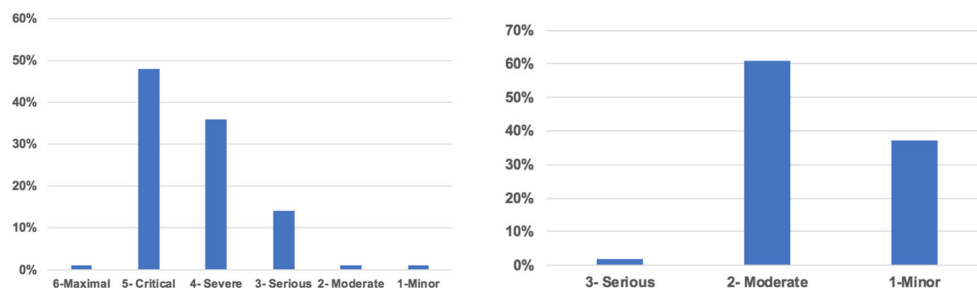


Fig. 4 Distribution of hard tissue injuries by type

Fig. 5 Comparison of injury severity for head injuries (left) and facial injuries (right)



neck injuries. Yet, when our cohort was also analysed for maxillofacial injuries only, a prevalence of 21% was noted, still higher than what has been reported thus far.

Many previous studies have demonstrated that men tend to be the victims of head, face or neck trauma more commonly than women [9, 10, 12, 13, 18–24]. This finding can also be seen in the present study which shows that major trauma patients with head, face or neck injuries were more likely to be male (M:F 2.125:1).

There are several hypotheses to explain this pattern including cultural attitudes towards occupation [20], social activities [9], hazardous driving [25] and the urban location of major trauma centres [22]. Though the reason(s) in the case of our study is unclear, it is interesting to note that the pattern persists.

Trauma has been shown to exert a significant health burden on all populations. It has however been traditionally viewed as a disease of the young, and in particular, of males involved in RTAs and interpersonal violence [26]. In recent times, major trauma in the elderly has become an increasingly recognised and studied phenomenon [27]. This is in line with our study finding that the mean age of injury was in the 5th decade. This is considerably higher than the age patterns demonstrated in the literature on craniofacial and neck injury which demonstrate mean ages ranging from 25 to 39 [9, 10, 18–23].

In many countries, the elderly comprise the most rapidly expanding section of the population. Indeed in Ireland, the increase in the number of persons over the age of 65 is approaching 20,000 per year [28]. This increase in the proportion of the population that is elderly, combined with increased life expectancy and activity levels all serve to explain the increasing numbers of elderly patients suffering major trauma.

Table 4 Length of stay distribution

Length of stay	No. of persons	% of persons
1 week or less	2580	48
Between 1 week and 1 month	1988	37
Between 1 month and 3 months	598	11
3 months or greater	198	4
Total	5364	100

This finding has important implications. Head, face and neck injuries are inherently associated with considerable morbidity. This is further worsened in the elderly owing to pre-existing disease, increasing frailty with age and slower recovery [29].

In our study, intoxication of some form was found in almost 30% of cases, with the most commonly implicated agent being alcohol. Furthermore, our study showed that the vast majority of injuries (84%) were non-intentional. There is a well-established link between alcohol and many types of unintentional injuries, RTAs, falls, burns and occupational injuries [30]. The link between alcohol consumption and facial injury specifically has also been documented in the literature. A 1998 UK survey found that at least 22% of all facial injuries were associated with alcohol consumption [11].

This underscores the continuingly important role of addressing alcohol consumption in the prevention of these injuries.

Historically, RTAs were the most common causes of both major trauma and craniofacial injury. This can be seen in studies carried out before the advent of modern advancements in vehicle safety, where up to 70% of injuries were caused by RTAs [13].

In recent times, however, there has been a significant shift in the aetiology of head and neck injury towards falls, and in particular, ‘low falls,’ i.e. those less than 2 m [24, 27]. Our study showed that falls were the most frequent cause of injury in almost two-thirds of cases, with low falls accounting for over half of all cases.

This pattern was seen in all age groups but the 0–35 group, where it was the second most common cause (Fig. 2).

This is supported by the finding that almost twice the number of injuries occurred at home (2322), compared with that on the roadside (1177). A large proportion of injuries due to falls occur in the elderly, and many occur during daily activities [31]. Consideration should therefore be given both to better home design to prevent this, as well as more widespread use of fall prevention strategies such as those outlined by Kannus et al. [32].

The causes of head and neck injury in this study were diverse, as were the resultant injuries. There was a total of 1272 counts of soft tissue injury, most frequently affecting the scalp or the face, and 5814 counts of bony fractures, most

commonly affecting the skull. The literature is conflicting with regard to the most commonly injured structures in the head and neck. Many studies on both major and non-major trauma reported the mandible as the most commonly injured structure [9, 20, 21], while in the others, it was the midface [13, 18]. Direct comparison of injury sites is complicated by the heterogeneity in anatomical areas included for analysis by studies in the literature.

Within the sample, head injuries tended to be both more prevalent and more severe than facial injuries, and this pattern was consistent across different age groups and genders.

Our study had an operative intervention rate of 26% (1402). This value is lower than the 43% seen in major trauma patients reported in the 2018 Irish Major Trauma Audit (MTA) [2] but higher than that seen in major trauma patients when considering maxillofacial injury alone [12].

Median length of stay in hospital in our study was 8 days with 90% of patients being discharged within a month.

The mortality rate in our study was 10%, twice the overall value reported in the MTA [2]. This is most probably due to the higher likelihood of injury to the brain and other critical structures that accompany a head and neck injury.

Conclusions

Head, face and neck injury occurs in over 1 in 4 Irish major trauma patients. It affects patients of all ages and sexes and produces considerable injury and surgical workload.

This study provides the first detailed description of the patterns head and neck injury in an Irish major trauma population. Its findings are reflective of a shift in the sociodemographic and aetiological landscape of major trauma in that the elderly are increasingly affected with more and more cases attributable to falls.

At a time where the Irish trauma system is seeing noteworthy advancements in its development, such as the release of the Major Trauma Audit, understanding the aetiology and epidemiology of these injuries is a crucial step in informing targeted preventative measures to reduce trauma burden. It also allows for efficient planning of care for these patients, and ensuring that this care meets both best-practice criteria as well as their needs.

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Availability of data and material The data that support the findings of this study are available from the Trauma Audit Research Network (TARN). Restrictions apply to the availability of these data, which were used under licence for this study. Data are available from the authors with the permission of the National Office of Clinical Audit.

Author's contributions All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by AM, JM and JC. The first draft of the manuscript was written by AM, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Compliance with ethical standards

Conflicts of interest/competing interests The authors declare that they have no conflicts of interest.

Ethics approval This study received ethical approval from the Clinical Research Ethics Committee of University College Cork.

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